Grade 7 Creative Arts and Design

Strand 1: Design

Lesson 2 — Drawing, Shading, Colouring and Modelling for Design

Codes: B7.1.2 (Drawing, Shading, Colouring and Modelling for Design) **Format:** Theory Questions and Answers

Preamble (pre-lesson)

Lesson 1 taught you to **see** design. This lesson teaches you to **show** it, using four skills:

1. **Drawing** — lines and shapes on paper.
2. **Shading** — light and dark so forms look solid.
3. **Colouring** — using colour to give mood and meaning.
4. **Modelling** — building a 3D form.

Words to know first:

- **Drawing** — making pictures with lines and marks.
- **Shading** — adding light and dark tones.
- **Tone** — how light or dark something looks.
- **Highlight** — the brightest part of a form.
- **Shadow** — the darkest part of a form.
- **Primary colours** — red, yellow, and blue.
- **Secondary colours** — orange, green, and purple.
- **Warm colours** — red, orange, and yellow.
- **Cool colours** — blue, green, and purple.
- **Modelling** — making a 3D form you can hold.
- **Texture** — how a surface feels or looks.

A. Drawing

Q1. What is drawing? **Ans.** Drawing is making pictures using lines and marks.

Q2. Which three elements do you use in drawing? **Ans.** You use dots, lines, and shapes in drawing.

Q3. What is the first step in a design drawing? **Ans.** The first step is to draw a light outline of the big shapes.

Q4. What do you do after drawing the light outline? **Ans.** You add smaller shapes and lines for details.

Q5. How do you make a drawing look solid? **Ans.** You make a drawing look solid by adding shading.

Q6. What is a curved line used for in drawing? **Ans.** A curved line is used for leaves, faces, and bowls.

Q7. What is a straight line used for in drawing? **Ans.** A straight line is used for edges, roads, and stems.

B. Shading

Q8. What is shading? **Ans.** Shading is adding light and dark tones to a drawing.

Q9. What is tone? **Ans.** Tone is how light or dark a part of a drawing looks.

Q10. What is a highlight? **Ans.** A highlight is the brightest part of a form where the light hits it.

Q11. What is a shadow? **Ans.** A shadow is the darkest part of a form away from the light.

Q12. Mention four shading techniques. **Ans.**

- Hatching
- Cross-hatching
- Stippling
- Smooth shading

Q13. What is hatching? **Ans.** Hatching is shading using a set of parallel lines.

Q14. What is cross-hatching? **Ans.** Cross-hatching is shading using two sets of lines that cross each other.

Q15. What is stippling? **Ans.** Stippling is shading using many small dots.

Q16. What is smooth shading? **Ans.** Smooth shading is shading done by blending pencil strokes until they look even.

Q17. Which part of a sphere is the highlight? **Ans.** The top side of the sphere facing the light is the highlight.

Q18. Which part of a sphere is the shadow? **Ans.** The bottom side of the sphere away from the light is the shadow.

C. Colouring

Q19. What are the primary colours? **Ans.** The primary colours are red, yellow, and blue.

Q20. What are the secondary colours? **Ans.** The secondary colours are orange, green, and purple.

Q21. What colour do you get when you mix red and yellow? **Ans.** You get orange when you mix red and yellow.

Q22. What colour do you get when you mix yellow and blue? **Ans.** You get green when you mix yellow and blue.

Q23. What colour do you get when you mix red and blue? **Ans.** You get purple when you mix red and blue.

Q24. What are warm colours? **Ans.** Warm colours are red, orange, and yellow.

Q25. What are cool colours? **Ans.** Cool colours are blue, green, and purple.

Q26. What mood do warm colours give? **Ans.** Warm colours give a mood of energy, heat, and closeness.

Q27. What mood do cool colours give? **Ans.** Cool colours give a mood of calm, distance, and coolness.

Q28. How does repeating a colour help a design? **Ans.** Repeating a colour makes the design look united and whole.

Q29. How does contrast help a design? **Ans.** Contrast makes one part of a design stand out from the rest.

Q30. Which colour would you use for a "hot sun" poster? **Ans.** I would use red or orange for a "hot sun" poster because they are warm colours.

Q31. Which colour would you use for a "calm lake" poster? **Ans.** I would use blue for a "calm lake" poster because it is a cool colour.

D. Modelling

Q32. What is modelling? **Ans.** Modelling is making a three-dimensional form you can hold.

Q33. Mention four materials used in modelling. **Ans.**

- Clay
- Cardboard
- Paper
- Wire

Q34. Which material is best for organic forms? **Ans.** Clay is best for organic forms such as faces and animals.

Q35. Which material is best for boxes and buildings? **Ans.** Cardboard is best for boxes and buildings.

Q36. What is the first step in modelling? **Ans.** The first step is to plan the model with a quick sketch.

Q37. Why do you build the main shape first? **Ans.** You build the main shape first so the model stands and has a strong structure.

Q38. How do you make a model stand? **Ans.** You balance the form so it stands without falling.

Q39. What is texture in modelling? **Ans.** Texture is how a surface feels or looks, such as rough or smooth.

Q40. How is folding paper used in modelling? **Ans.** Folding paper is used to make cubes, pyramids, and simple structures.

Text diagram — shading on a sphere

```
light
\
v
( . . ) <-- highlight (brightest)
( ... )
( ..... )
( .... ) <-- mid-tone
( .. ) <-- shadow (darkest)
```

Text diagram — colour wheel

```
YELLOW
/ \
ORANGE GREEN
\ /
RED —— BLUE
|
PURPLE
```

Text diagram — warm and cool

```
WARM (energy) COOL (calm)
red blue
orange green
yellow purple
```

Exercises (8 sections × 5 questions)

Exercise 1 — Drawing

1. What is drawing?
2. Which three elements do you use in drawing?
3. What is the first step in a design drawing?
4. How do you make a drawing look solid?
5. **[Review]** What is a shape?

Answers

1. Drawing is making pictures using lines and marks.
2. Dots, lines, and shapes.
3. The first step is to draw a light outline of the big shapes.
4. You make a drawing look solid by adding shading.
5. A shape is a closed outline you can see.

Exercise 2 — Shading

1. What is shading?
2. What is a highlight?
3. What is a shadow?
4. Mention three shading techniques.
5. **[Review]** What is tone?

Answers

1. Shading is adding light and dark tones to a drawing.
2. A highlight is the brightest part of a form where the light hits it.
3. A shadow is the darkest part of a form away from the light.
4. Hatching, cross-hatching, stippling.
5. Tone is how light or dark a part of a drawing looks.

Exercise 3 — Shading techniques

1. What is hatching?
2. What is cross-hatching?
3. What is stippling?
4. What is smooth shading?
5. **[Review]** Name the three elements of design.

Answers

1. Hatching is shading using a set of parallel lines.
2. Cross-hatching is shading using two sets of lines that cross each other.
3. Stippling is shading using many small dots.
4. Smooth shading is shading done by blending pencil strokes until they look even.
5. Dots, lines, and shapes.

Exercise 4 — Colour wheel

1. What are the primary colours?
2. What are the secondary colours?
3. What colour do you get when you mix red and yellow?
4. What colour do you get when you mix yellow and blue?
5. **[Review]** What is balance?

Answers

1. The primary colours are red, yellow, and blue.
2. The secondary colours are orange, green, and purple.
3. You get orange when you mix red and yellow.
4. You get green when you mix yellow and blue.
5. Balance is when the parts of a design feel evenly arranged.

Exercise 5 — Warm and cool colours

1. What are warm colours?
2. What are cool colours?
3. What mood do warm colours give?
4. What mood do cool colours give?
5. **[Review]** What is rhythm?

Answers

1. Warm colours are red, orange, and yellow.
2. Cool colours are blue, green, and purple.
3. Warm colours give a mood of energy, heat, and closeness.
4. Cool colours give a mood of calm, distance, and coolness.
5. Rhythm is a sense of movement created by repeating elements.

Exercise 6 — Using colour in design

1. How does repeating a colour help a design?
2. How does contrast help a design?

3. Which colour would you use for a "hot sun" poster?
4. Which colour would you use for a "calm lake" poster?
5. **[Review]** What is repetition?

Answers

1. Repeating a colour makes the design look united and whole.
2. Contrast makes one part of a design stand out from the rest.
3. I would use red or orange because they are warm colours.
4. I would use blue because it is a cool colour.
5. Repetition is using the same element again and again.

Exercise 7 — Modelling

1. What is modelling?
2. Mention four materials used in modelling.
3. Which material is best for organic forms?
4. What is the first step in modelling?
5. **[Review]** What is texture?

Answers

1. Modelling is making a three-dimensional form you can hold.
2. Clay, cardboard, paper, wire.
3. Clay is best for organic forms.
4. The first step is to plan the model with a quick sketch.
5. Texture is how a surface feels or looks.

Exercise 8 — Project and scenario

1. Design a simple book cover using one element and one principle.
2. How do you shade a sphere?
3. How do you make a cube from paper?
4. **[Review]** What is biomimicry?
5. **[Review]** What are the four skills in this lesson?

Answers

1. Use dots as a repeated pattern and arrange the title in balance with the picture.
2. You leave a bright highlight on the light side and darken the side away from the light.
3. You fold a paper net of six squares and glue the edges together.
4. Biomimicry is copying ideas from nature to solve human problems.
5. Drawing, shading, colouring, and modelling.

Summary table

Term	One-line meaning
Drawing	making pictures with lines and marks
Shading	adding light and dark tones
Primary colours	red, yellow, blue
Warm / cool	warm = R/O/Y; cool = B/G/P
Modelling	making a 3D form you can hold

NaCCA CCP B7.1.2 — Theory Questions and Answers format. Plain simple English. British notation.